CHAPTER 1

Digital System Basics

Analog vs digital · logic levels · positive/negative logic · timing, fan-in/fan-out, noise margin, power

1.1 What is a digital system?

● LEVEL 1 · BEGINNER

What is it? A digital system stores and processes information using only **two values**, which we write as **0** and **1**. Everything a computer does (text, music, photos, videos) is finally stored as long lists of 0s and 1s.

Why only two values? Electronic circuits find it easy to make and to tell apart two clearly different voltages, for example “low” and “high”. If a little electrical disturbance (noise) changes the voltage slightly, the circuit can still tell which of the two values was meant.

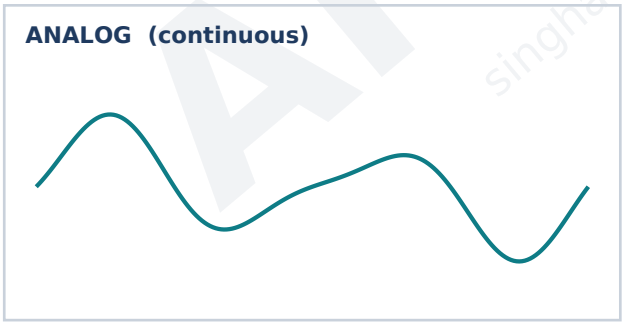
Real-life analogy. A light switch has two states, OFF and ON. Nobody is confused if the switch wobbles a little; it is still clearly OFF or ON. A dimmer knob, on the other hand, has endless positions, so small errors change the brightness.

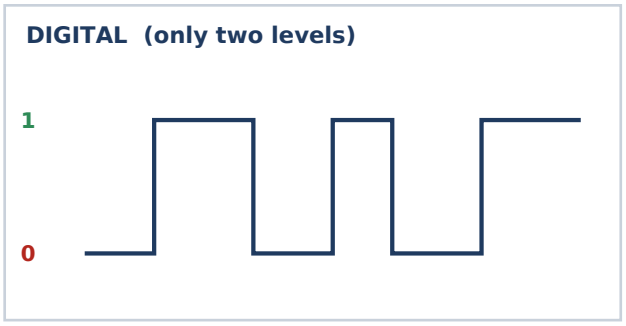
```
Switch OFF  ->  0  ->  LOW  voltage
Switch ON   ->  1  ->  HIGH voltage
```

1.2 Analog vs Digital

● LEVEL 1 · BEGINNER

An **analog** signal can take **any value** in a range (like room temperature: 20.1°C, 20.15°C, 20.156°C...). A **digital** signal takes only a **fixed set of separate values** (mostly two).





	Analog	Digital
Values	Continuous, endless values	Discrete (separate) values, mostly 0 and 1
Effect of small noise	Changes the value (error stays)	Usually ignored (value is restored)
Examples	Sound wave, temperature, voltage from a microphone	Data inside a computer, a text file
Storing / copying	Quality falls each time	Perfect copies are possible

EASY MEMORY
Analog = continuous. Digital = discrete.

1.3 Logic levels, Logic 0 and Logic 1

● LEVEL 1 · BEGINNER

A digital circuit does not use one exact voltage for 1 and another for 0. It treats a **range** of voltages as **HIGH** and another range as **LOW**. The voltages in between are **forbidden**: the circuit does not promise what will happen there. For a 5 V TTL gate:

Voltage range	Meaning
0 V to 0.8 V	Accepted as LOW
0.8 V to 2.0 V	Forbidden (undefined) region
2.0 V to 5 V	Accepted as HIGH

Logic 0 and **Logic 1** are just names for these two ranges. Which range is called 0 and which is called 1 is our choice:

System	HIGH voltage means	LOW voltage means
Positive logic (normal)	Logic 1	Logic 0
Negative logic	Logic 0	Logic 1

Worked example: the same gate, two names

A gate gives a HIGH output only when both inputs are HIGH. Look at what it does in the two conventions:

A (volt.)	B (volt.)	Y (volt.)	Positive logic A B Y	Negative logic A B Y
L	L	L	0 0 0	1 1 1
L	H	L	0 1 0	1 0 1
H	L	L	1 0 0	0 1 1
H	H	H	1 1 1	0 0 0

Positive-logic columns show an **AND** gate (Y = 1 only for 1,1). Negative-logic columns show Y = 0 only for 0,0, which is an **OR** gate.

> **GATE POINT**
> The same physical gate is **AND in positive logic and OR in negative logic**. In the same way NAND ↔ NOR. This is called the **duality** of logic gates.

1.4 Bit, nibble, byte, word

● LEVEL 1 · BEGINNER

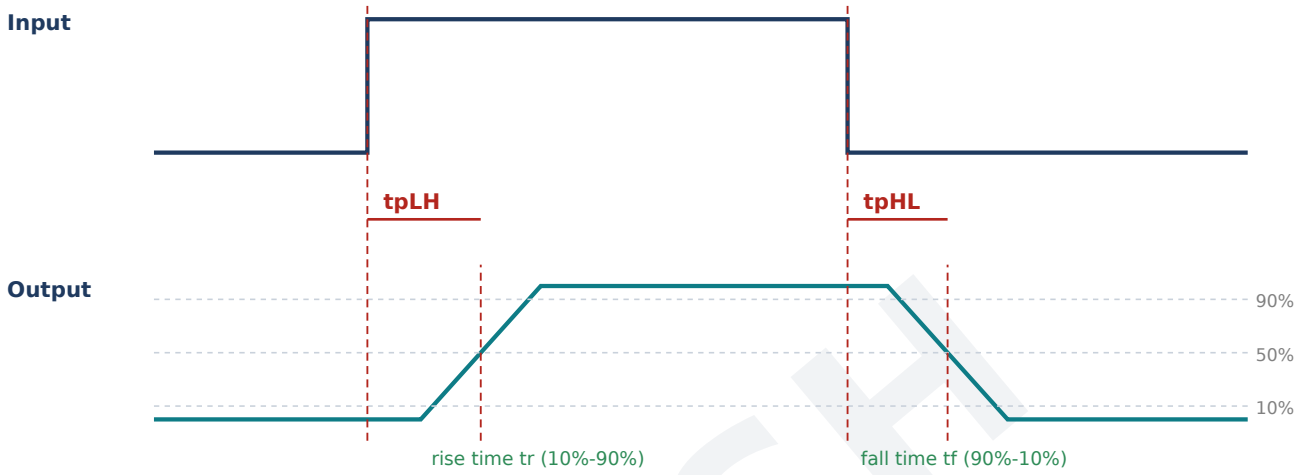
Name	Size	How many different values?
Bit	1 binary digit	2 (0 or 1)
Nibble	4 bits	$2^4 = 16$
Byte	8 bits	$2^8 = 256$
Word	Machine dependent (16, 32, 64 bits…)	2^n for an n-bit word

Rule: with n bits you can make $\mathbf{2^n}$ different patterns. *Why?* Each extra bit doubles the number of patterns: 1 bit → 2, 2 bits → 4, 3 bits → 8 …

1.5 Timing terms: propagation delay, rise time, fall time

● LEVEL 2 · EXAM

No circuit reacts instantly. When the input changes, the output changes *a little later*, and the change itself takes some time.



- **Propagation delay** means the small time a gate needs to change its output after its input changes. It is measured between the **50% points** of input and output.
- $\mathbf{t_{pLH}}$ = delay when the output goes LOW → HIGH. $\mathbf{t_{pHL}}$ = delay when the output goes HIGH → LOW.
- **Average propagation delay: $t_{pd} = (t_{pLH} + t_{pHL}) / 2$**
- **Rise time (t_r)**: time for the output to rise from **10% to 90%** of its final value.
- **Fall time (t_f)**: time to fall from **90% to 10%**.

Worked example 1 (Level 2)

Given: t_{pLH} = 12 ns, t_{pHL} = 8 ns.

Concept: average of the two delays.

Calculation: t_{pd} = (12 + 8) / 2 = 10 ns.

Answer: 10 ns.

GATE POINT

Delays add up. If signals pass through gates one after another, total delay = sum of the gate delays. This is why a ripple-carry adder or a ripple counter is slow (later chapters).

1.6 Fan-in and fan-out

● LEVEL 2 · EXAM

- **Fan-in** = the number of inputs of a gate. A 3-input NAND gate has fan-in 3.
- **Fan-out** = the maximum number of gate inputs that one gate output can drive without going wrong. It exists because every connected input draws a little current from the output.

```
                +--> input of gate 2
Gate 1 output --+--> input of gate 3     Fan-out of gate 1 = 3 here
                +--> input of gate 4
```

Formula. An output can *supply* (HIGH) or *absorb* (LOW) only a limited current. Each input needs a fixed current. So:

```
Fan-out (HIGH state) = I_OH(max) / I_IH          I_OH = output HIGH current (max)
Fan-out (LOW  state) = I_OL(max) / I_IL          I_IH = input HIGH current
Fan-out              = the SMALLER of the two    I_OL, I_IL = same for LOW state
```

Worked example 2 (Level 3)

Given: I_{OH} = 400 µA, I_{IH} = 40 µA, I_{OL} = 8 mA, I_{IL} = 1.6 mA.

Concept: find fan-out for each state; the circuit must work in both, so take the smaller.

Calculation: HIGH: 400 / 40 = 10. LOW: 8 / 1.6 = 5.

Answer: fan-out = min(10, 5) = **5**.

EXAM TRAP
Do not quote only one state. The **smaller** value is the fan-out.

1.7 Noise margin

● **LEVEL 2 · EXAM**

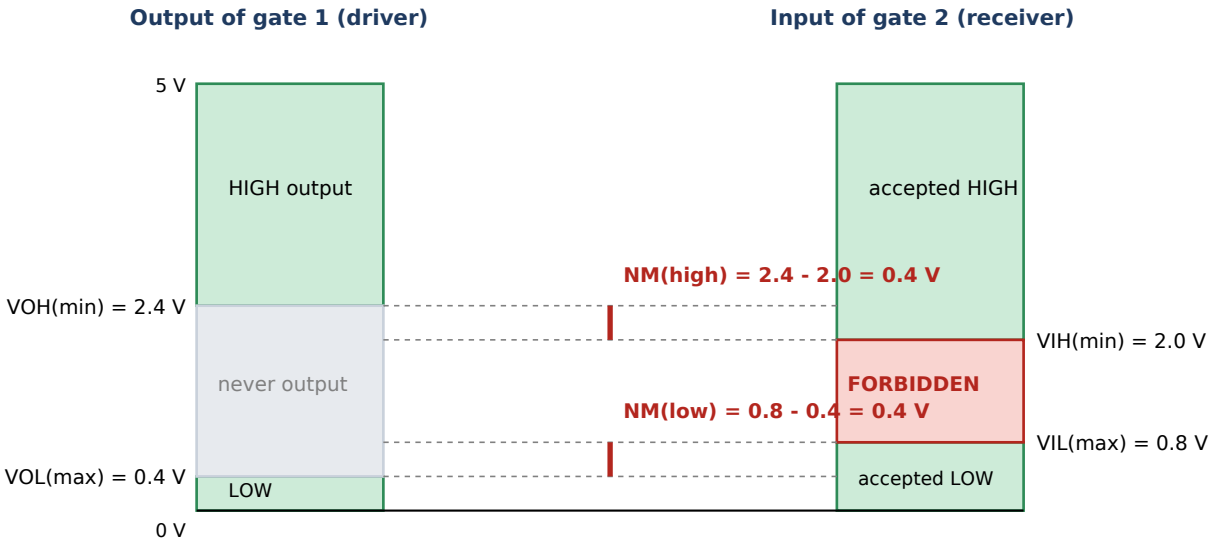
Noise means unwanted small voltage disturbances on a wire. **Noise margin** tells us how much noise a signal can carry and still be understood correctly by the next gate.

Think of a driver aiming for the middle of a road (the *output* level) and the road having a wide safe area (the *input* requirement). The gap between them is your safety margin.

Four voltages describe a gate family:

Symbol	Meaning
V_{OH}(min)	Lowest voltage a gate guarantees at its HIGH output
V_{OL}(max)	Highest voltage a gate guarantees at its LOW output
V_{IH}(min)	Lowest input voltage the next gate still accepts as HIGH
V_{IL}(max)	Highest input voltage the next gate still accepts as LOW

```
NM_HIGH = V_OH(min) - V_IH(min)          (output guarantee  minus  input requirement)
NM_LOW  = V_IL(max) - V_OL(max)
```



The overall noise margin of the family is the **smaller** of NM_{HIGH} and NM_{LOW}.

Worked example 3 (Level 2)

Given: V_{OH}(min) = 2.7 V, V_{IH}(min) = 2.0 V, V_{IL}(max) = 0.8 V, V_{OL}(max) = 0.5 V.

Calculation: NM_{HIGH} = 2.7 − 2.0 = 0.7 V. NM_{LOW} = 0.8 − 0.5 = 0.3 V.

Answer: NM_{HIGH} = 0.7 V, NM_{LOW} = 0.3 V, family noise margin = **0.3 V**.

COMMON MISTAKE

Mixing up the subtraction order. Remember: **output guarantee minus input requirement** for HIGH, and **input limit minus output guarantee** for LOW. A noise margin can never be negative in a working family.

1.8 Power dissipation and speed-power product

● **LEVEL 2 · EXAM**

- **Power dissipation** = electrical power a gate uses (and turns into heat): $\mathbf{P = V_{CC} \times I_{CC}}$. Unit: mW (or µW).
- **Speed-power product (SPP)** = $\mathbf{t_{pd} \times P}$. Unit: **pico-joule (pJ)** (ns × mW = pJ). **Smaller is better.**

Why do we need SPP? Speed and power pull in opposite directions: a very fast gate usually burns more power. SPP gives one number to compare gates fairly.

Worked example 4 (Level 2)

Given: Gate A: t_{pd} = 10 ns, P = 10 mW. Gate B: t_{pd} = 20 ns, P = 2 mW.

Calculation: SPP(A) = 10 × 10 = 100 pJ. SPP(B) = 20 × 2 = 40 pJ.

Answer: Gate A is faster, but gate B has the better (smaller) speed-power product.

1.9 Exam points and common mistakes

GATE POINT

Ring oscillator. An odd number n of inverters joined in a ring oscillates. **Period = 2 × n × t_{pd}**. *Why:* a change must travel round the ring twice (once to make output HIGH, once to make it LOW). Example: n = 5, t_{pd} = 3 ns → period = 30 ns → frequency = 33.3 MHz.

EXAM TRAP

Rise time and fall time use **10%–90%** points. Propagation delay uses **50%** points. Do not mix them.

COMMON MISTAKE

Thinking logic 1 is always 5 V. Logic levels depend on the family and supply voltage (TTL 5 V, CMOS 3.3 V, 1.8 V…). And in negative logic HIGH means 0.

GATE QUICK CHECK — Chapter 1

Formulas

```
t_pd = (t_pLH + t_pHL) / 2            Values with n bits = 2^n
NM_H = V_OH(min) - V_IH(min)          NM_L = V_IL(max) - V_OL(max)
Fan-out = min( I_OH/I_IH , I_OL/I_IL ) SPP = t_pd x P   (pJ)
Ring oscillator period = 2 x n x t_pd  (n odd)
```

Facts to remember: analog = continuous, digital = discrete · nibble = 4 bits, byte = 8 bits · AND in positive logic = OR in negative logic · smaller SPP is better.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Which is a digital signal: (a) the voltage from a microphone, (b) the data bits stored in a pen drive, (c) room temperature?

Q2. How many different values can be stored in 1 byte? How many bits are there in 3 nibbles?

Q3. In negative logic, what does a HIGH voltage represent?

● **LEVEL 2 · EXAM**

Q4. A gate has t_{pLH} = 15 ns and t_{pHL} = 9 ns. Find t_{pd}.

Q5. A positive-logic NAND gate is used in negative logic. Which gate is it now?

Q6. V_{OH}(min) = 3.6 V, V_{IH}(min) = 2.2 V, V_{IL}(max) = 1.0 V, V_{OL}(max) = 0.4 V. Find both noise margins.

● **LEVEL 3 · GATE**

Q7. A gate output can source 600 µA and sink 12 mA. Each input needs 20 µA (HIGH) and 1.5 mA (LOW). What is the fan-out?

Q8. Gate P: t_{pd} = 5 ns, 20 mW. Gate Q: t_{pd} = 25 ns, 1 mW. Which is faster? Which has the better speed-power product?

Q9. A ring of 7 inverters, each with t_{pd} = 2 ns. Find the oscillation frequency.

Q10. An output rises from 0 V to 5 V. It crosses 0.5 V at t = 3 ns and 4.5 V at t = 11 ns. Find the rise time. If it takes t_{pd} = 6 ns to cross 2.5 V after the input change at t = 0, what is t_{pLH} for this edge?

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Answers

Q	Solution (Given → Concept → Calculation → Answer)
1	(b). Stored bits take only 0/1. (a) and (c) vary continuously.
2	Byte: 2^8 = 256 values. 3 nibbles = 3 × 4 = 12 bits.
3	Logic 0 (in negative logic HIGH = 0, LOW = 1).
4	t_{pd} = (15 + 9)/2 = **12 ns**.
5	Positive NAND = negative **NOR** (duality).
6	NM_H = 3.6 − 2.2 = **1.4 V**. NM_L = 1.0 − 0.4 = **0.6 V**.
7	HIGH: 600/20 = 30. LOW: 12 mA / 1.5 mA = 8. Fan-out = min = **8**.
8	SPP(P) = 5 × 20 = 100 pJ. SPP(Q) = 25 × 1 = 25 pJ. **P is faster**, **Q has the better SPP**.
9	Period = 2 × 7 × 2 ns = 28 ns. f = 1/28 ns = **35.7 MHz**.
10	t_r = 11 − 3 = **8 ns** (10%–90% points). The delay to the 50% point (2.5 V) is given as 6 ns, so t_{pLH} = **6 ns** (propagation delay uses the 50% level, not 10%–90%).

Next: Chapter 2 — Number Systems & Codes (binary, octal, hex, fractions, BCD, Excess-3, Gray, signed numbers).

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